

Wyatt Walsh

SENIOR AI/ML ENGINEER

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Summary: Senior AI/ML engineer shipping production agentic AI systems, LLM document intelligence, RAG/retrieval workflows, AI guardrails, and validated data pipelines and platforms for fintech, risk, compliance, and enterprise tooling.

EXPERIENCE

Senior AI/ML Engineer

Frame Payments | Remote | Oct 2025 - Jan 2026

Delivered production agentic AI document intelligence, human-in-the-loop review, and multimodal compliance screening for high risk merchant onboarding.

- Deployed production LLM document intelligence with PydanticAI, Amazon Bedrock AgentCore, and Claude; added human-in-the-loop browser feedback for confidential uploads and reduced analyst follow-ups/resubmissions ~90% by blocking miscategorized documents before review.
- Built multimodal restricted-item compliance screening by combining product images and structured metadata; translated banking-partner policy requirements into automated risk checks for merchant onboarding across tens of merchants and multi-location catalogs.
- Built a NotebookLM-based technical knowledge hub spanning vendor/web documentation and internal monorepo docs; automated corpus refresh with a Colab pipeline that re-ingests the latest sources to keep retrieval accurate and up to date.
- Drove internal AI adoption with Notion MCP playbooks, high-signal prompt patterns, and weekly office hours reaching ~15% of the company; standardized PII/confidential-document safeguards and auditability requirements.

Associate III, Senior Software Engineer - Data Integrations

JPMorgan Chase & Co. | New York, NY | May 2023 - Apr 2025

Built enterprise data pipelines, risk analytics, and data-quality automation for credit trading while scaling secure internal AI developer tooling.

- Pioneered firm-wide rollout of secure generative AI developer tools (LLMs, GitHub Copilot, Codeium) as one of the first 500 engineers, shaping scalable deployment and leading workshops for technical and non-technical teams.
- Devised ML-driven data-quality anomaly detection for credit trading and risk analytics, reducing false positives by 25% and cutting manual review time so risk teams could focus on critical alerts.
- Developed resilient Python wrapper for KDB+, democratizing data access for 5,000+ users with CI-style tests and optimized Q functions, reducing manual processing by over 10 hours per week.
- Engineered a cloud-function-driven asset-fee data sink, consolidating 10+ sources into a centralized database, cutting manual maintenance by 90% and providing on-demand access for traders through a Python SDK, REST API, and interactive dashboard.

Senior Software Engineer

Stealth Web3 Startup | Remote | Mar 2022 - Jul 2022

- Architected a scalable collateralized lending application using React, Next.js, TypeScript, MobX, and ethers.js, translating complex wallet and smart-contract flows into secure, performant user workflows.
- Implemented end-to-end CI/CD with GitHub Actions, Husky, Cypress, and Jest to halve release cycles and raise test coverage to 90%, while mentoring a front-end intern to build key UI components.
- Optimized front-end performance through code splitting, lazy loading, and asset caching, reducing initial load times by 40% and JavaScript bundle size by 30%.

Founder, Chief Technology Officer

SandLabs | Remote | Jun 2021 - Mar 2022

- Engineered and launched a Next.js/TailwindCSS public site with top-tier Lighthouse scores and secured \$75K in grant funding via a competitive ecosystem program.
- Engineered Python pipelines to extract and normalize a 1M+ Reddit posts/comments dataset and publish a reusable data product for external analytics workflows.

Demand Forecasting Extern

Gap Inc. | San Francisco, CA | Jan 2020 - May 2020

- Applied time-series decomposition and bias correction techniques to demand forecasts, boosting accuracy by up to 25% and minimizing surplus inventory costs.
- Owned the data engineering layer by architecting scalable MySQL and Apache Spark pipelines across 5+ years of sales data, improving data processing throughput by 95% and standardizing insights across forecasting teams.

SKILLS

MACHINE LEARNING AND GENERATIVE AI: scikit-learn, Hugging Face Transformers, OpenAI API, LangChain, LangGraph, Model Context Protocol (MCP), Retrieval Augmented Generation (RAG), Prompt Engineering, PydanticAI, Amazon Bedrock, Claude, GitHub Copilot, Document AI, AI Guardrails

DATA ENGINEERING AND ANALYTICS: Pandas, NumPy, SQLAlchemy, Pydantic, Pandera, Apache Spark, AWS Glue, KDB+, PostgreSQL, DynamoDB, ETL, DuckDB, Polars, Parquet

PROGRAMMING & SCRIPTING LANGUAGES: Python, SQL, TypeScript, JavaScript, Go, Rust, Bash

CLOUD INFRASTRUCTURE AND DEVOPS: AWS, Docker, Kubernetes, Terraform, GitHub Actions, AWS Lambda, Google Cloud Functions, ECS, EKS

PROJECTS

AI Agent Harness Configs | github.com/wyattowalsh/agents

LLM developer-workflow catalog with 40+ reusable skills, MCP-ready configs, shared instructions, and cross-runtime agent automation.

- Built 40+ reusable instruction and skill packs plus shared harness configs that install across Claude Code, Codex, GitHub Copilot, Cursor, and related agent runtimes.

Tech: Python, TypeScript, MCP, FastMCP

NBA Basketball Database | github.com/wyattowalsh/nbadb

Star-schema analytics warehouse spanning 65K+ NBA games and 13M+ play-by-play events with validated ETL, DuckDB/SQLite/Parquet exports, docs, CLI, and Kaggle distribution.

- Implemented a high-performance, Pandera-validated ETL pipeline that extracts from the NBA API, updates warehouse outputs daily, and exports DuckDB, SQLite, Parquet, and CSV artifacts.
- Built docs, schema references, release workflows, and Kaggle distribution for a 180+ table public warehouse spanning 130+ stats.nba.com endpoint classes.

Tech: Python, Pandas, Polars, SQLAlchemy, DuckDB, Pandera

ProxyWhirl | github.com/wyattowalsh/proxywhirl

Data-collection toolkit with nine rotation strategies, 100+ auto-validated sources, and self-healing proxy health checks.

- Implemented 9 rotation strategies (round-robin, weighted, geo-targeted, performance-based, and more) with configurable fallback chains and per-source health scoring.

Tech: Python, SQLite, PyPI

AI/ML Web Feeds | github.com/wyattowalsh/ai-web-feeds

Typed AI/ML source-discovery platform with HTTP/RSS validation, contributor workflows, OPML exports, and docs-backed feeds for model and agent research.

- Defined a structured YAML feed schema with topic classification (ML, NLP, CV, etc.) and source-type validation (blog, paper, newsletter) to ensure feed quality at scale.

Tech: Python, Pydantic, SQLAlchemy, YAML, Fumadocs, Vercel

MCP-Crawl4AI | github.com/wyattowalsh/mcp-crawl4ai

FastMCP server enabling AI agents to crawl, extract, and return structured web content over documented stdio/HTTP paths.

- Built a FastMCP v3 server with lifespan-managed headless Chromium, session-aware crawling workflows, and automatic Playwright browser setup—only 2 runtime dependencies.

Tech: Python, FastMCP, Crawl4AI, Playwright, Docker

MDXPad | github.com/wyattowalsh/mdxpad

Native macOS MDX editor with synchronized preview, multi-provider AI assistance, and Keychain-backed credential storage.

- Built a dual-format editor on Electron 39 + CodeMirror 6 with synchronized scroll between source and rendered preview, remark/rehype plugin support, and YAML frontmatter validation.

Tech: Electron, React, TypeScript, CodeMirror, Vercel AI SDK

FL Studio MCP Server | github.com/wyattowalsh/fl-mcp

Alpha MCP platform for AI-assisted music production with 190+ typed FL Studio tools, resources-first contracts, transaction-safe mutations, and 650+ tests.

- Defined 190+ FL Studio tools plus core governance tools, with read paths exposed through typed resources and mutations funneled through a transaction pipeline.

Tech: Python, MCP, Next.js, Fumadocs, FL Studio SDK

Personal Website: w4w.dev | github.com/wyattowalsh/personal-website

Long-running personal web app and MDX publishing system pairing a polished frontend with durable content, SEO, and authoring workflows.

- Designed and maintained the public product surface across multiple generations, including responsive landing experiences, theme support, motion, and accessibility-conscious UI polish.

Tech: Next.js, React, TypeScript, Tailwind CSS, MDX

EDUCATION & CERTIFICATIONS

Bachelor of Science

University of California, Berkeley - College of Engineering, Industrial Engineering & Operations Research
Aug 2014 - Dec 2020

AWS Certified Cloud Practitioner

Amazon Web Services (AWS) | Nov 2024

High School Diploma

The Hotchkiss School
Aug 2012 - May 2014

IBM Data Science Professional Certificate

IBM | Feb 2021